Pattern Card 1 — Incorrect Example Output

Event ID(s): a1c12e6d4

Short name: Incorrect example output

What happened in code: StringUtils.abbreviate() produces an abbreviated string according to its existing implementation.

What happened in doc: A Javadoc example showed an incorrect output and was subsequently corrected.

Why the doc is wrong: The documented example gives callers an incorrect expectation of the method's result.

Smallest trigger: Change the expected output of one Javadoc example to a string different from the actual return value.

Pattern Card 2 — Incorrect Nullability Documentation

Event ID(s): 285cd2fc4

Short name: Incorrect parameter nullability

What happened in code: setAccessibleTrue() removed handling for null arguments, requiring a non-null parameter.

What happened in doc: The old Javadoc allowed null; the correction required a non-null value.

Why the doc is wrong: A caller could pass null based on the documentation even though the implementation no longer handles it.

Smallest trigger: Change “must not be null” to “may be null” while the code requires a non-null argument.

Pattern Card 3 — Outdated Parameter Type or Name

Event ID(s): 003 (7e5906fb0)

Short name: Stale parameter type and name

What happened in code: The parameter of setAccessibleTrue() changed from Field field to AccessibleObject accessibleObject.

What happened in doc: The Javadoc was updated to describe the generalized parameter instead of the old Field parameter.

Why the doc is wrong: The old description incorrectly narrows the accepted type and uses an outdated parameter name.

Smallest trigger: Replace the current parameter type or name in the Javadoc with an outdated one after a signature change.

Pattern Card 4 — Incorrect Boundary Condition

Event ID(s): 005

Short name: Inclusive versus exclusive boundary

What happened in code: LocalVariableTable.getLocalVariable() changed its end condition from pc <= endPc to pc < endPc.

What happened in doc: The Javadoc was unchanged and did not explicitly state whether the endpoint was inclusive.

Why the doc is wrong: A claim that the endpoint is inclusive would contradict the updated code. **This is a proposed mutation pattern, not a confirmed error in the unchanged Javadoc.**

Smallest trigger: Change a correct boundary claim from pc < endPc to pc <= endPc, or “exclusive” to “inclusive.”

Pattern Card 5 — Missing Exception Documentation

Event ID(s): 006, 007, 009

Short name: Missing @throws for invalid input

What happened in code: Methods reject invalid values by throwing ClassGenException. Event 007 introduced increment bounds validation, event 009 added an upper bound on nargs, and event 006 already contained bounds validation.

What happened in doc: Events 006 and 007 added @throws ClassGenException documentation; event 009 left Javadoc unchanged despite its new validation.

Why the doc is wrong: Callers are not told about an exception they may encounter for invalid arguments. This is a documentation-omission pattern rather than necessarily a false affirmative claim.

Smallest trigger: Delete the existing @throws ClassGenException tag while retaining the throwing behavior.

Pattern Card 6 — Incorrect Parameter Range Restriction

Event ID(s): 006, 007, 009

Short name: Incorrect accepted parameter range

What happened in code: The methods enforce numeric limits; for example, INVOKEINTERFACE requires nargs from 1 through Const.MAX_BYTE.

What happened in doc: Parameter descriptions do not always explicitly include the enforced range.

Why the doc is wrong: An affirmative claim that values outside the permitted range are accepted would conflict with the implementation. **This is a candidate mutation, not proof that the original generic descriptions were wrong.**

Smallest trigger: Widen a correct documented range such as 1 <= nargs <= Const.MAX_BYTE to nargs >= 1 with no upper limit.

Pattern Card 7 — Incorrect Instruction Length

Event ID(s): 008, 010

Short name: Stale instruction-length documentation

What happened in code: Event 008 corrected length handling for wide local-variable instructions; event 010 explicitly set INVOKEDYNAMIC length to 5 bytes.

What happened in doc: The related Javadocs were unchanged and did not explicitly state the corrected lengths.

Why the doc is wrong: A false length claim would mislead developers computing bytecode offsets. **The unchanged generic comments were not confirmed wrong.**

Smallest trigger: Change a correct documented instruction length, such as 5 bytes for INVOKEDYNAMIC, to 3 bytes.

Pattern Card 8 — Incorrect Type Validation

Event ID(s): 011, 012, 013

Short name: Incorrect accepted array type

What happened in code: The verifier changed array-element validation: DALOAD checks the immediate element type, while CALOAD specifically requires a char array.

What happened in doc: The unchanged Javadocs generically say that instruction preconditions are checked.

Why the doc is wrong: A claim that more array types are accepted than the implementation permits would be false. **The existing generic Javadocs are not themselves confirmed contradictions.**

Smallest trigger: Change “requires a char array” to “accepts any array type” for CALOAD.

Pattern Card 9 — Incorrect Return-Value Description

Event ID(s): 015 (3475afc43aae34001dc45f1697dcf3d389ebb9c4)

Short name: Incorrect return value for special inputs

What happened in code: checkNotNull() was updated to return an empty string for non-string values instead of casting them to String.

What happened in doc: New Javadoc explained that null values and certain internal attributes of other object types yield an empty string.

Why the doc is wrong: Claiming that non-string values are cast or returned as strings contradicts the updated implementation.

Smallest trigger: Change “returns an empty string for non-string values” to “casts non-string values to String.”

Pattern Card 10 — Missing Reference to Overloaded Method

Event ID(s): 017

Short name: Missing related-method reference

What happened in code: An outerHtml(Appendable) overload became available while the existing outerHtml() implementation remained unchanged.

What happened in doc: A @see #outerHtml(Appendable) reference was added.

Why the doc is wrong: Without the reference, the Javadoc is less complete, but it is not necessarily factually false.

Smallest trigger: Remove the @see #outerHtml(Appendable) tag.